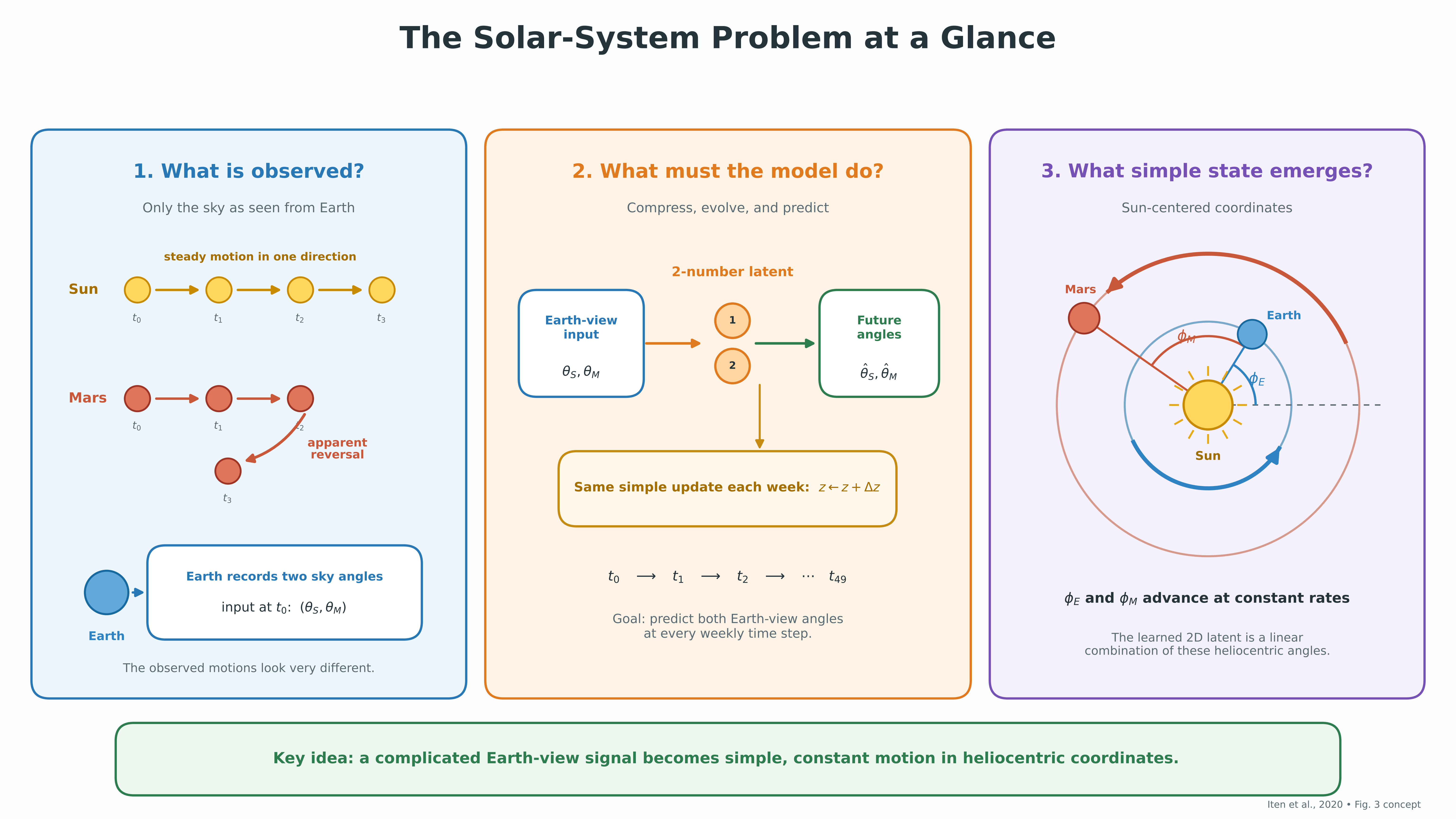
The Solar-System Problem at a Glance
1. What is observed?
Only the sky as seen from Earth
steady motion in one direction
Sun
t_0
t_1
t_2
t_3
Mars
t_0
t_1
t_2
apparent reversal
t_3
Earth records two sky angles
input at t_0: (θ_S, θ_M)
Earth
The observed motions look very different.
2. What must the model do?
Compress, evolve, and predict
2-number latent
Earth-view input
θ_S, θ_M
1
2
Future angles
$\hat{\theta}_S, \hat{\theta}_M$
Same simple update each week: $z \leftarrow z + \Delta z$
$t_0 \longrightarrow t_1 \longrightarrow t_2 \longrightarrow \cdots\ t_{49}$
Goal: predict both Earth-view angles at every weekly time step.
3. What simple state emerges?
Sun-centered coordinates
Mars
Earth
ϕ_M
ϕ_E
Sun
ϕ_E and ϕ_M advance at constant rates
The learned 2D latent is a linear combination of these heliocentric angles.
Key idea: a complicated Earth-view signal becomes simple, constant motion in heliocentric coordinates.
Iten et al., 2020 • Fig. 3 concept